COCOMO II Technical Valuation Report (2025)

Scope
- Repository: ‘capibaraGPT_v3‘
- Objective: Internal technical valuation (non-accounting) using a calibrated COCOMO II scenario.
- Date: 2026-02-07

Baseline Inputs
- ‘KLOC‘: ‘128.28‘
- ‘A‘: ‘0.35‘ (internally calibrated for high AI-assisted productivity)
- ‘Cost per person-month‘: ‘5,500‘ (same currency used in the calculation)

Scale Factors (SF)
- ‘PREC=1.24‘
- ‘FLEX=1.01‘
- ‘RESL=2.83‘
- ‘TEAM=2.19‘
- ‘PMAT=3.12‘

Derived:
- ‘B = 0.91 + 0.01 * sum(SF) = 1.0139‘

Effort Multipliers (EM)
- ‘RELY=1.00‘
- ‘DATA=1.00‘
- ‘CPLX=1.00‘
- ‘RUSE=1.00‘
- ‘TIME=1.00‘
- ‘PVOL=1.00‘
- ‘ACAP=0.85‘
- ‘PCAP=0.88‘

Derived:
- ‘EAF = product(EM) = 0.7480‘

COCOMO II Calculation
Formulas:
- ‘Effort (PM) = A * KLOC^B * EAF‘
- ‘D = 0.28 + 0.2 * (B - 0.91)‘
- ‘Schedule (months) = 3.67 * Effort^D‘
- ‘Total Cost = Effort * Cost per PM‘

Computed values:
- ‘Effort‘: ‘35.927914‘ person-months
- ‘Schedule‘: ‘10.777273‘ months
- ‘Total Cost‘: ‘197,603.527182‘

Interpretation
- This is a **technical valuation scenario** calibrated to a low-effort assumption (‘A=0.35‘).
- It is not a canonical default COCOMO II calibration.
- Use this report as internal reference for technical valuation alignment.

2024 vs 2025 Comparison
Because repository history does not contain commits from 2024, 2024 LOC cannot be measured directly from ‘git‘.
The 2024 LOC shown below is an inferred estimate using the 2025 ratio:

- ‘2025 EUR/LOC = 197,603.53 / 128,280 = 1.5406 EUR/LOC‘
- ‘2024 inferred LOC = 115,000 / 1.5406 = 74,646 LOC‘ (approx.)

Year	LOC	KLOC	Technical valuation (EUR)	Source quality
2024	74,646 (inferred)	74.65	115,000.00	Estimated (no 2024 git snapshot)
2025	128,280 (measured)	128.28	197,603.53	Measured from current repo + COCOMO II (‘A=0.35‘)

Reproducibility
Command used:

```bash
python COCOMO_II/cocomo_ii.py --kloc 128.28 --cost-per-person-month 5500 --a 0.35
```

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